```python
# # 1. Install the package
# !git clone https://github.com/JKCooper2/gym-bandits.git
# %cd gym-bandits
# !pip install -e .

# 2. Import Libraries
import gym
import gym_bandits
import numpy as np
import matplotlib.pyplot as plt

# 3. Create Environment
# We typically wrap this to handle deprecated warnings or API changes if necessary,
# but here we use the standard call.
env = gym.make("BanditTenArmedGaussian-v0")

# 4. Define the Agent
class EpsilonGreedyAgent:
    def __init__(self, epsilon, n_arms):
        self.epsilon = epsilon
        self.n_arms = n_arms
        self.q_values = np.zeros(n_arms)
        self.action_counts = np.zeros(n_arms)

    def select_action(self):
        # Exploration: Random arm
        if np.random.rand() < self.epsilon:
            return np.random.choice(self.n_arms)
        # Exploitation: Best known arm
        else:
            # Use random tie-breaking
            return np.random.choice(np.flatnonzero(self.q_values == self.q_values.max()))

    def update_q_values(self, action, reward):
        self.action_counts[action] += 1
        # Q(a) = Q(a) + 1/N(a) * (R - Q(a))
        self.q_values[action] += (reward - self.q_values[action]) / self.action_counts[actio

# 5. Initialize parameters
epsilon_greedy_agent = EpsilonGreedyAgent(epsilon=0.1, n_arms=env.action_space.n)
num_steps = 1000  # Number of pulls

# 6. Training Loop
print(f"Training Epsilon Greedy (Eps=0.1) on {env.spec.id}...")
rewards = []
```

```python
# Note: In Bandits, we typically treat the problem as 1 long episode of N steps,
# rather than N episodes of 1 step.
env.reset()

for step in range(num_steps):
    action = epsilon_greedy_agent.select_action()

    # Gym API Handling (Handles New vs Old Gym versions)
    step_result = env.step(action)

    if len(step_result) == 5:
        # New Gym API: obs, reward, terminated, truncated, info
        _, reward, terminated, truncated, _ = step_result
    else:
        # Old Gym API: obs, reward, done, info
        _, reward, done, _ = step_result

    epsilon_greedy_agent.update_q_values(action, reward)
    rewards.append(reward)

# 7. Results
print("\nEstimated Q-Values (Agent's belief):")
print(epsilon_greedy_agent.q_values)
print(f"Average Reward: {np.mean(rewards):.4f}")
```

```
Gym has been unmaintained since 2022 and does not support NumPy 2.0 amongst other critical f
Please upgrade to Gymnasium, the maintained drop-in replacement of Gym, or contact the autho
Users of this version of Gym should be able to simply replace 'import gym' with 'import gymn
See the migration guide at https://gymnasium.farama.org/introduction/migration_guide/ for ad
```

```
Training Epsilon Greedy (Eps=0.1) on BanditTenArmedGaussian-v0...

Estimated Q-Values (Agent's belief):
[ 1.2450885   1.41697565 -2.07561285 -0.41306027 -1.18806099 -1.56609936
  0.8519358  -1.1566144   1.50183809  2.76472256]
Average Reward: 2.3894
```

```
c:\Python311\Lib\site-packages\gym\utils\passive_env_checker.py:174: UserWarning: WARN: Futu
  logger.warn(
c:\Python311\Lib\site-packages\gym\utils\passive_env_checker.py:187: UserWarning: WARN: Futu
  logger.warn(
c:\Python311\Lib\site-packages\gym\utils\passive_env_checker.py:195: UserWarning: WARN: The
  logger.warn(
c:\Python311\Lib\site-packages\gym\utils\passive_env_checker.py:219: DeprecationWarning: WAR
  logger.deprecation(
```

```python
import numpy as np
import matplotlib.pyplot as plt
```

```python
import gym
import gym_bandits

# --- 1. Define Agents ---

class EpsilonGreedyAgent:
    def __init__(self, n_arms, epsilon=0.1):
        self.n_arms = n_arms
        self.epsilon = epsilon
        self.q_values = np.zeros(n_arms)
        self.counts = np.zeros(n_arms)

    def select_action(self):
        if np.random.rand() < self.epsilon:
            return np.random.choice(self.n_arms)
        return np.argmax(self.q_values)

    def update(self, action, reward):
        self.counts[action] += 1
        self.q_values[action] += (reward - self.q_values[action]) / self.counts[action]

class UCBAgent:
    def __init__(self, n_arms, c=2):
        self.n_arms = n_arms
        self.c = c
        self.q_values = np.zeros(n_arms)
        self.counts = np.zeros(n_arms)
        self.t = 0

    def select_action(self):
        self.t += 1
        if 0 in self.counts:
            return np.argmin(self.counts)
        uncertainty = self.c * np.sqrt(np.log(self.t) / self.counts)
        return np.argmax(self.q_values + uncertainty)

    def update(self, action, reward):
        self.counts[action] += 1
        self.q_values[action] += (reward - self.q_values[action]) / self.counts[action]

# Note: Standard Thompson Sampling uses Beta distributions for Bernoulli rewards (0 or 1).
# Since "BanditTenArmedGaussian" returns continuous rewards (Gaussian),
# we use Gaussian Thompson Sampling (Normal-Inverse-Gamma prior is ideal, but we will simpl
class GaussianThompsonSamplingAgent:
    def __init__(self, n_arms):
        self.n_arms = n_arms
```

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        # Prior: Mean=0, Precision (tau)=0.0001 (High uncertainty)
        self.mu = np.zeros(n_arms)
        self.tau = np.zeros(n_arms) + 0.0001

    def select_action(self):
        # Sample from Normal(mu, 1/tau)
        samples = np.random.normal(self.mu, 1 / np.sqrt(self.tau))
        return np.argmax(samples)

    def update(self, action, reward):
        # Bayesian update for Gaussian with known variance (assumed 1)
        # precision_new = precision_old + 1
        # mean_new = (precision_old * mean_old + reward) / precision_new
        old_tau = self.tau[action]
        self.tau[action] += 1
        self.mu[action] = (old_tau * self.mu[action] + reward) / self.tau[action]

# --- 2. Simulation ---

def run_experiment(agent_class, n_arms, steps, **kwargs):
    env = gym.make("BanditTenArmedGaussian-v0")
    env.reset()
    agent = agent_class(n_arms, **kwargs)
    rewards = []

    for _ in range(steps):
        action = agent.select_action()
        step_result = env.step(action)

        # Handle API
        if len(step_result) == 5:
            _, reward, _, _, _ = step_result
        else:
            _, reward, _, _ = step_result

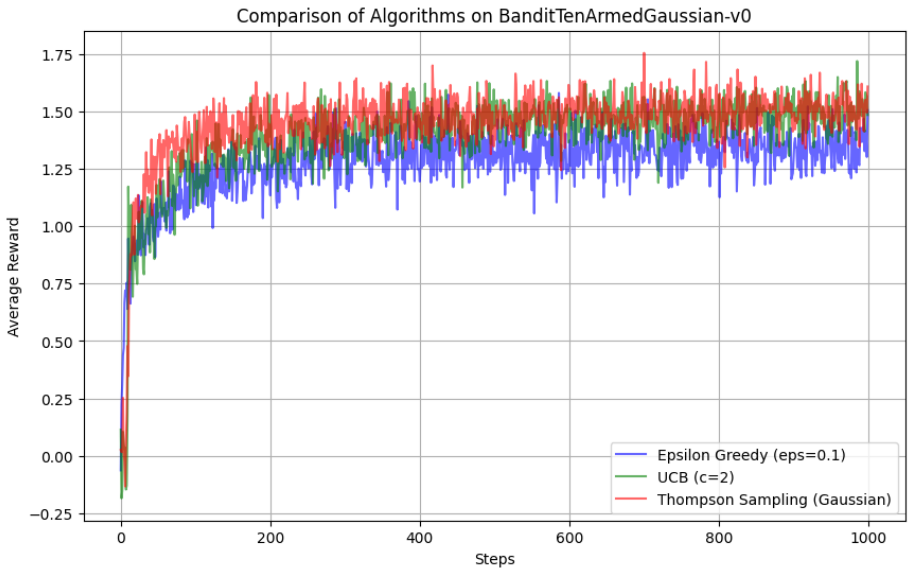
        agent.update(action, reward)
        rewards.append(reward)
    return np.array(rewards)

# Parameters
steps = 1000
runs = 200 # Average over 200 runs for smooth lines
n_arms = 10

print("Running Comparison Simulation...")
```

```python
# Storage for averaging
avg_r_eps = np.zeros(steps)
avg_r_ucb = np.zeros(steps)
avg_r_ts = np.zeros(steps)

for r in range(runs):
    avg_r_eps += run_experiment(EpsilonGreedyAgent, n_arms, steps, epsilon=0.1)
    avg_r_ucb += run_experiment(UCBAgent, n_arms, steps, c=2)
    avg_r_ts += run_experiment(GaussianThompsonSamplingAgent, n_arms, steps)

avg_r_eps /= runs
avg_r_ucb /= runs
avg_r_ts /= runs

# --- 3. Plotting ---
plt.figure(figsize=(10, 6))
plt.plot(avg_r_eps, label='Epsilon Greedy (eps=0.1)', color='blue', alpha=0.6)
plt.plot(avg_r_ucb, label='UCB (c=2)', color='green', alpha=0.6)
plt.plot(avg_r_ts, label='Thompson Sampling (Gaussian)', color='red', alpha=0.6)

plt.xlabel('Steps')
plt.ylabel('Average Reward')
plt.title('Comparison of Algorithms on BanditTenArmedGaussian-v0')
plt.legend()
plt.grid(True)
plt.show()
```

```
Running Comparison Simulation...
```


Comparison of Algorithms on BanditTenArmedGaussian-v0
Average Reward
Steps
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
0
200
400
600
800
1000
Epsilon Greedy (eps=0.1)
UCB (c=2)
Thompson Sampling (Gaussian)